Animesh Biswas

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EDUCATION

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, Odisha
B. Tech, Electronics and Telecommunication Engineering
CGPA: 8.41/10.0 | Expected Graduation: May 2027

Jamshedpur Public School, Jamshedpur, Jharkhand
Higher Secondary Certificate (CBSE Class XII) | Completed: March 2023

RESEARCH PUBLICATIONS

Design and Implementation of a High-Performance 16-Bit DSP ALU Using Six-Stage Pipelining | *IEEE IC2PCT (Accepted for Presentation, May 15-17)*
Tools: Xilinx Vivado Design Suite, Verilog HDL, Digilent Nexys 4 DDR (Xilinx Artix-7)

- Developed a 6-stage pipelined architecture on Nexys 4 DDR, optimizing throughput by explicitly inferring DSP48E1 hard macros for multiplication.
- Achieved 100 MHz operational frequency and low dynamic power (25mW) by implementing latency balancing and register retiming strategies.
- Authored as 2nd author for the 2026 4th International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES).
- Code Repository: GitHub Repository

RESEARCH EXPERIENCE

Minor Research: Development and Implementation of Proportionate-Type & VLSI Algorithms (Ongoing) | KIIT
Tools: MATLAB, HDL Converter (MATLAB), Verilog HDL, Xilinx Vivado, FPGA

- Simulating proportionate-type adaptive algorithms for system identification to analyze convergence and performance baselines in MATLAB.
- Developing the VLSI architecture by translating high-level MATLAB models into synthesizable Verilog HDL for hardware acceleration on FPGA.

PROJECTS

Autonomous Sand Rover | *Kshitij, IIT Kharagpur (Jan 2025)*
Tools: Autodesk Fusion 360, Arduino (C), Motor Drivers (L293D), ESP32, ESCs

- Engineered the drive system using Electronic Speed Controllers (ESCs) and Arduino C to achieve smoother, precise manoeuvrability in sand terrain.
- Integrated a camera module for low-light navigation and upgraded the mechanical chassis with improved traction for autonomous cave exploration.

RFID Attendance System Using Raspberry Pi Pico W | Electronics Design Lab, KIIT (2026)
Tools: Raspberry Pi Pico W, MicroPython, Flask, Docker, SQLite, RFID (RC522)

- Developed a full-stack IoT system using Raspberry Pi Pico W and Flask to transmit scanned RFID UIDs over Wi-Fi for real-time processing and SQLite database storage.
- Designed a secure, Dockerized web dashboard featuring live monitoring, responsive UI, and automated Excel export functionality for attendance records.

TECHNICAL SKILLS

Languages: C, Python, Verilog, MATLAB
Software: KiCad, Xilinx Vivado, VS Code, Arduino IDE, MATLAB
Hardware: Microcontrollers (AVR, ESP32, Raspberry Pi Zero, Raspberry Pi W), FPGA, PCB Design & Prototyping
Domains: Analog & Digital Circuit, Power Electronics, Semiconductor Devices